Ahmed Emad

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI Engineer with a data science background and a BSc in Artificial Intelligence, skilled in Python, SQL, and machine learning. Experienced in building RAG systems, LLM-powered applications, and multi-agent workflows, and in serving them through FastAPI REST APIs. Strong foundation in statistics, NLP, and data analysis with Power BI.

EDUCATION

Bachelor of Science in Artificial Intelligence Oct 2021 – Jun 2025
Menoufia University, Egypt

EXPERIENCE

AI Engineer Intern | 3S Smart Solutions Jun 2026 – Sep 2026

- Designed and implemented an AI-powered HR chatbot on Odoo ERP enabling employees to instantly query HR information through natural language, reducing dependency on human HR intervention.
- Built a multi-agent pipeline using CrewAI to handle intent recognition, data retrieval, and response formatting across HR-related queries.
- Developed and served the system through a FastAPI backend, integrating the agentic pipeline with Odoo's HR module for real-time data access.
- Engineered structured output handling and conversation flow to ensure accurate, context-aware responses to employee queries.

PROJECTS

Chat with your Documents— RAG Pipeline 2026

Python | FastAPI | Qdrant | sentence-transformers | LangChain | Streamlit | Docker Compose

- Built an end-to-end RAG pipeline with FastAPI: asynchronous PDF upload, background ingestion (extraction, chunking, embedding, indexing), and an LLM answer endpoint grounded in retrieved context with filename and page-number citations. Qdrant vector database containerized with Docker Compose.
- Implemented hybrid retrieval in Qdrant, combining multilingual dense embeddings (BAAI/bge-m3) with BM25 sparse vectors and fusing results with Reciprocal Rank Fusion to capture both semantic and exact-keyword matches.

YouTube Comments Analyzer — NLP Sentiment & Insight Tool 2026

Python | FastAPI | Hugging Face Transformers | YouTube Data API v3 | Pandas | Scikit-learn

- Engineered an end-to-end NLP pipeline to fetch, preprocess, and analyze YouTube comments at scale using the YouTube Data API v3, supporting batch retrieval of thousands of comments per video.
- Implemented multilingual sentiment analysis using a fine-tuned Hugging Face Transformer model, classifying comments into positive, negative, and neutral categories with confidence scoring.
- Built topic modeling and keyword extraction modules to surface dominant themes and frequently discussed subjects within comment sections, enabling actionable content insights.
- Developed a production-ready FastAPI backend exposing RESTful endpoints for comment ingestion, sentiment scoring, keyword extraction, and aggregated analytics reporting.
- Designed an interactive dashboard to visualize sentiment distributions, trending topics, and engagement patterns, providing creators and analysts with data-driven audience feedback.

TECHNICAL SKILLS

Programming Languages: Python, SQL
Machine Learning & AI: Supervised Learning, Unsupervised Learning, Deep Learning, Feature Engineering, Model Evaluation, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Sentiment Analysis, Topic Modeling
LLM & Agent Frameworks: LangChain, Hugging Face Transformers, CrewAI,
Python Libraries: NumPy, Pandas, Scikit-learn, PyTorch
Backend & API Development: FastAPI, RESTful API Design
Databases & Vector Stores: ChromaDB, Qdrant, Microsoft SQL Server
Data Visualization & BI: Tableau, Power BI, Excel
Mathematics & Statistics: Linear Algebra, Statistics, Hypothesis Testing
Tools & Version Control: Git, GitHub

CERTIFICATIONS

Retrieval Augmented Generation (RAG) |Supervised Machine Learning: Regression | Linear Algebra for Machine Learning and Data Science | Python for Data Science, AI & Development | Data Modeling in Power BI | Extract, Transform and Load Data in Power BI | Harnessing the Power of Data with Power BI | Introduction to Data Analytics

LANGUAGES

Arabic (Native/Bilingual) | English (Proficient)